Practical - 09

Write a program for Log Capturing and Event Correlation

Prerequisite:

- Latest version of Squid should be used.(version 2.5 or greater)
- A web server for testing purpose which can be used instead of Internet.
- Squid Version greater than 2.6 is required for Transparent squid proxy configuration in this lab.
-

Learning Objectives:

- To understand how Log Records are generated for Further Analysis.

New Concepts:

- Squid and Sarg

Theory

1.1. Introduction:

- During the period of development of internet, users are allowed for unlimited access to the resources due to less number of users. So there were less issues related to accessing speed over internet.
- With the increase in internet usage, many issues raised related to accessing speed, effective bandwidth utilization etc. One method of overcoming these issues is, maintaining a copy of webpage visited by a user in the cache so that the other user who visits the same webpage will access the same website within a short period of time. This method not only increases the accessing speed but also helps in utilizing the bandwidth effectively.
- The above said functionality can be achieved by maintaining a proxy server through which all the users in the organization or a group access the internet. The most widely used proxy server in Linux is Squid Proxy, which is free software released General Public License.

- Squid provides proxy and cache services for **H**yper **T**ext **T**ransfer **P**rotocol (HTTP),

  File **T**ransfer **P**rotocol (FTP), and various other protocols.

To configure a system as a proxy server, one should have a sufficient amount of memory for maintaining the cache which in turn increases the performance.

- In case if the internet connection is not available, setup one host as a web server in place of internet and assign the IP address to the proxy server network interface in the network, used by web server instead of public IP address assigned to that interface.

1.2. Steps to Configure Squid Proxy:

1.2.1. Installation of Squid Package

A Squid proxy server is generally installed on a separate server than the Web server with the original files. Squid works by tracking object use over the network. Squid will initially act as an intermediary, simply passing the client's request on to the server and saving a copy of the requested object. If the same client or multiple clients request the same object before it expires from Squid's cache, Squid can then immediately serve it, accelerating the download and saving bandwidth.

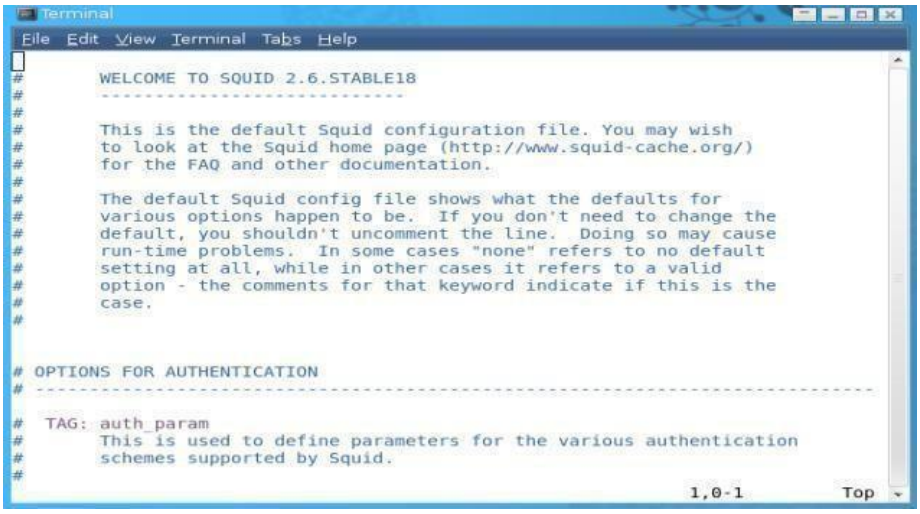
```
sudo apt update
sudo apt -y install squid
```

1.2.2. Accessing the Proxy Server configuration file

To configure squid proxy server we need to edit the *sudo gedit /etc/squid/squid.conf* file and the default location of squid.conf file varies from distribution to distribution and from version to version. We can edit the configuration file using vi editor through command prompt.

```
sudo gedit /etc/squid/squid.conf
```

Then the content of the configuration file can be viewed as shown below in the figure.



Editing the squid configuration file

sudo gedit /etc/squid/squid.conf

Search the TAG: auth_param and paste the following acl

```
auth_param basic program /usr/lib/squid/basic_ncsa_auth /etc/squid/passwd
auth_param basic children 5
auth_param basic realm Squid Basic Authentication
auth_param basic credentialsttl 2 hours
acl auth_users proxy_auth REQUIRED
http_access allow auth_users

 #search localnet and paste line in last

acl localnet src 192.168.60.70
```

>sudo service squid3 restart

Specifying the interface and port number on which the proxy server should listen.

By default, the proxy server will listen on all the available network interfaces on the system for requests. For Example, if one interface card is assigned a public ip from which it is connected to internet and the other interface card is assigned an ip address which belongs to your local area network. Then in order to make you proxy server to listen for requests from your Local Area Network through a particular port, then change the variable http_port 3128 in the squid configuration file to desired ip address and port number in the format shown below.

http_port <ip address belonging to LAN>:<port number>

Example: For example, if your proxy server has an ip address 192.168.60.70 which belongs to the local area network 192.168.60.0/24 and you want the server to listen for requests from your LAN through a particular port say 3456, then you can change the variable http_port as shown.

http_port 192.168.60.70:3456

Assigning Access Controls

By default, no user machine is allowed to connect to the proxy server except the localhost. To allow the local machines access your proxy server, locate the acl section in the squid configuration file starting with acl and at the end of the last acl line specify your access

control. For example to allow local area network 192.168.60.0/24 machines to access your proxy server, specify the acl as

acl mylan src 192.168.60.0/255.255.255.0

In the above example, mylan specifies the name of my access control. We can specify any name other than my lan for access control. src specifies the source network.

Allow or Deny based on Access Control.

After specifying the access control for your local LAN, we need to provide allow permission for the specified LAN using http_access variable in the squid configuration file as shown in the example below.

Example: To allow the above specified access control (i.e acl mylan src 192.168.60.0/255.255.255.0), we need to specify the http_access variable as

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Hyderabad http_access allow mylan

Here mylan specifies the access control used. Suppose if we want to allow all the networks except the 192.168.60.0/24 network to access the proxy then we can specify the http_access variable as

http_access deny !mylan

In the above line, !mylan specifies except mylan network.

Note:

The above specified http_access variable should be specified before the line

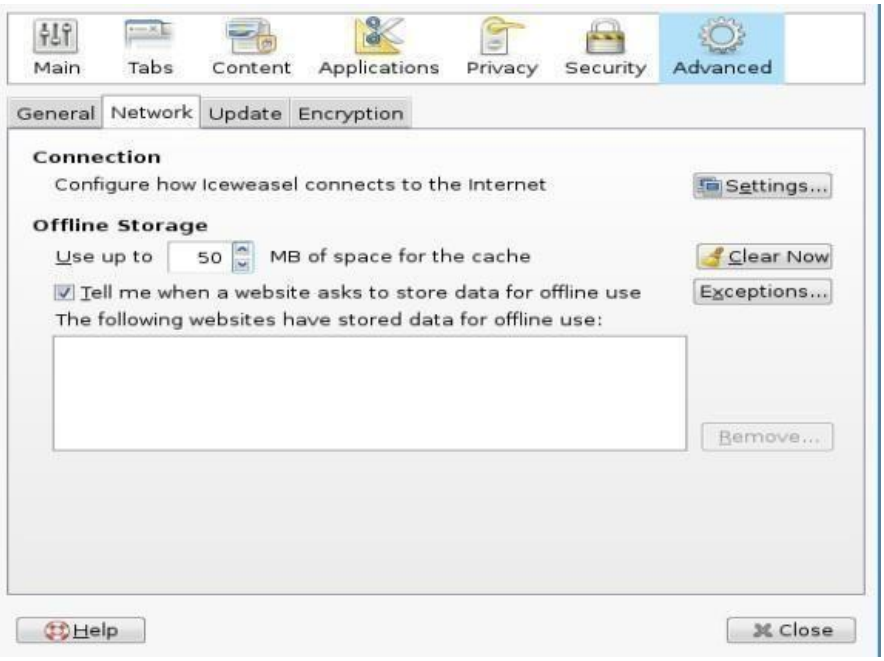
http_access deny all in the configuration file.

Saving the changes and exit the gedit Editor

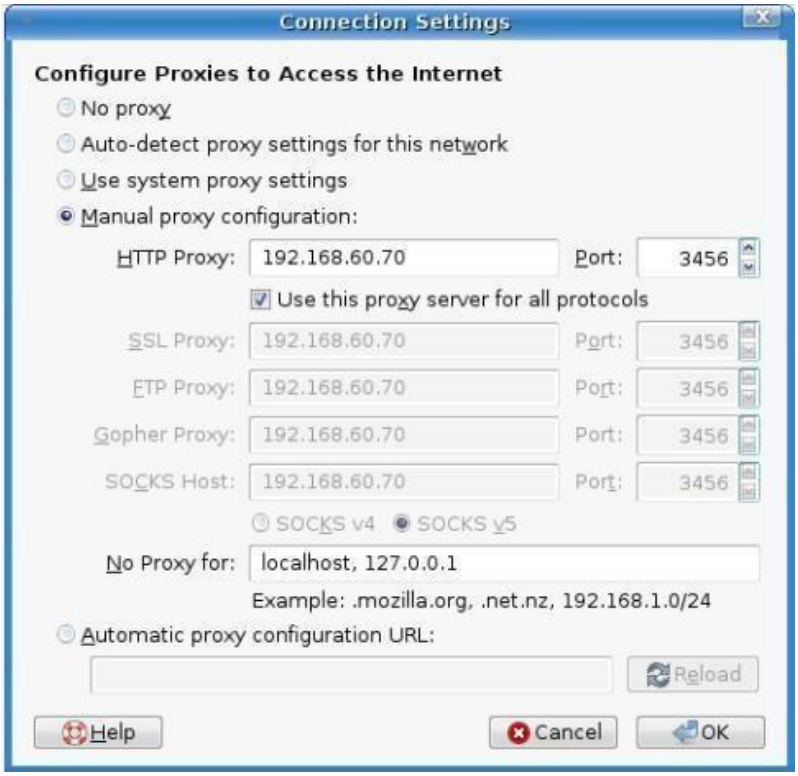
After making appropriate changes to your configuration file exit the vi editor window by pressing Esc followed by :wq!. Here wq specifies save changes and exit the configuration file.

4.3 Testing the Squid configuration

To test the squid configuration, open a browser in any one of the pc in local area network or on the proxy server and specify the proxy settings as the ipaddress of the proxy server and port on which it is listening for requests. For example, in firefox web browser if we want to set the proxy settings in the browser window goto **Edit** -->**Preferences** and window similar to shown below will be displayed.



Now select Advanced tab, and under advanced tab click on Network tab and click on Settings option under Connection field. Then a window similar to the shown below will be displayed.



4.4SARG – Squid Analysis Report Generator and Internet Bandwidth Monitoring Tool

SARG is an open source tool that allows you to analyze the squid log files and generates beautiful reports in HTML format with information's about users, IP addresses, top accessed sites, total bandwidth usage, elapsed time, downloads, access denied websites, daily reports, weekly reports and monthly reports.

The SARG is very handy tool to view how much internet bandwidth is utilized by individual machines on the network and can watch on which websites the network's users are accessing.

Installing Sarg from Source

The '**sarg**'package by default not included in **RedHat** based distributions, so we need to manually compile and install it from source tar ball. For this, we need some additional pre- requisites packages to be installed on the system before compiling it from source.

$ sudo apt-get install sarg

Configuring Sarg

Now it's time to edit some parameters in SARG main configuration file. The file contains lots of options to edit, but we will only edit required parameters like:

Access logs

path

Output

directory

Date

Format

Overwrite report for the same date.

Open sarg.conf file with your choice of editor and make changes as shown below.

vi /usr/local/etc/sarg.conf [On RedHat based systems]

Now Uncomment and add the original path to your squid access
log file. # sarg.conf

TAG: access_log file
Where is the access.log file
sarg
-l file
access_log /var/log/squid/access.log

Next, add the correct Output directory path to save the generate squid reports in that directory. Please note, under Debian based distributions the Apache web root directory is '/var/www'. So, please be careful while adding correct web root paths under your Linux distributions.

TAG: output_dir
The reports will be saved in that
directory # sarg -o dir

output_dir /var/www/html/squid-reports

Set the correct date format for reports. For example, 'date_format e' will

display reports in 'dd/mm/yy' format.

TAG: date_format

Date format in reports: e (European=dd/mm/yy), u (American=mm/dd/yy), w (Weekly=yy.ww)

#

date_format e

Next, uncomment and set Overwrite report to 'Yes'. # TAG: overwrite_report yes|no

yes - if report date already exist then will be overwritten.

no - if report date already exist then will be renamed to filename.n, filename.n+1

overwrite_report yes

That's it! Save and close the file.

Step 3: Generating Sarg Report

Once, you've done with the configuration part, it's time to generate the squid log report using the following command.

sarg -x [On RedHat based systems]

Assessing Sarg Report

The generated reports placed under '/var/www/html/squid-reports/' or '/var/www/squid-reports/' which can be accessed from the web browser using the address.

http://localhost/squid-reports OR

http://ip-address/squid-reports

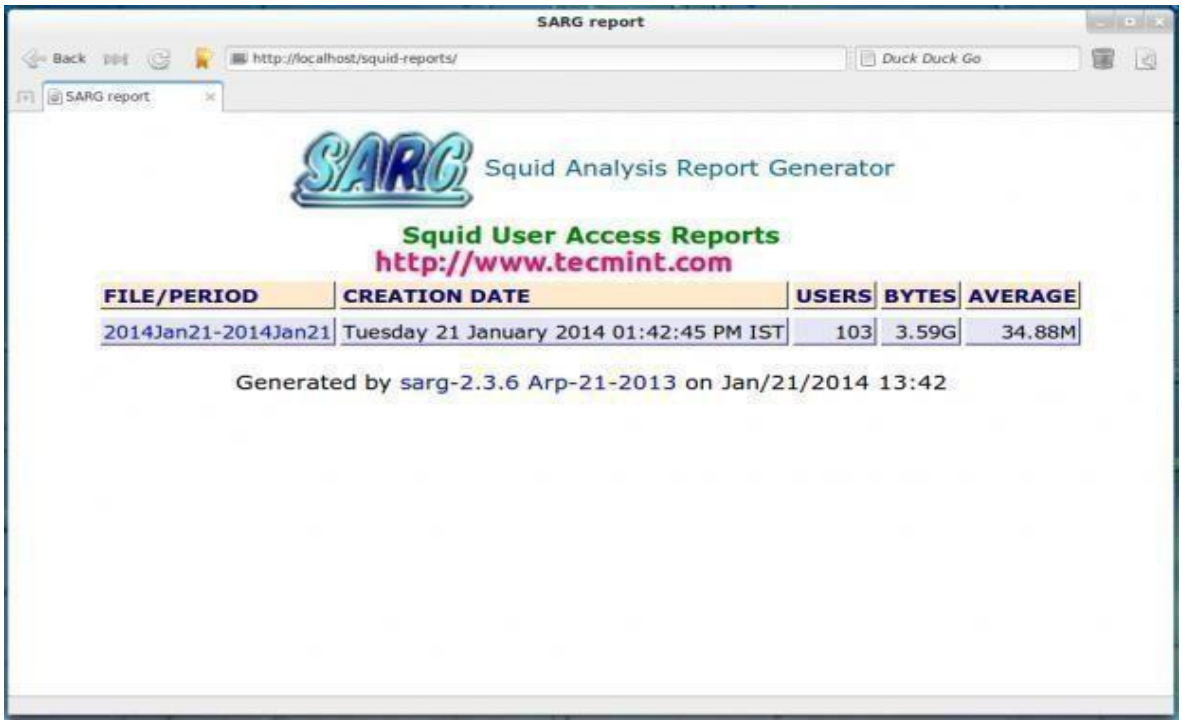


Fig.1 Sarg Main Window

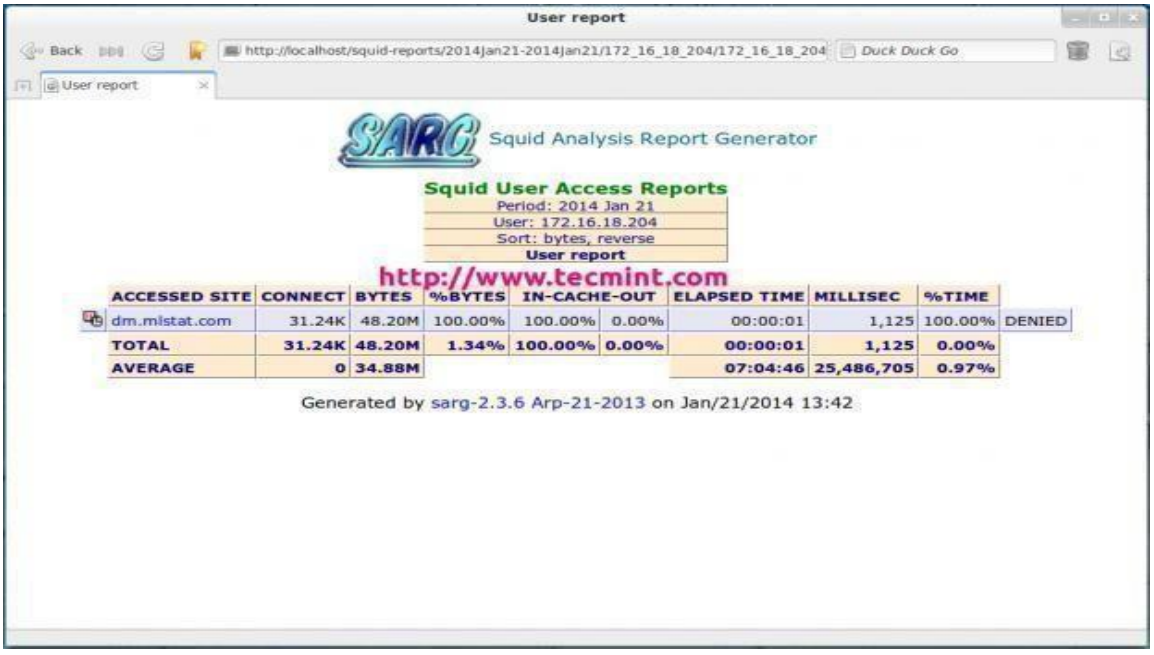
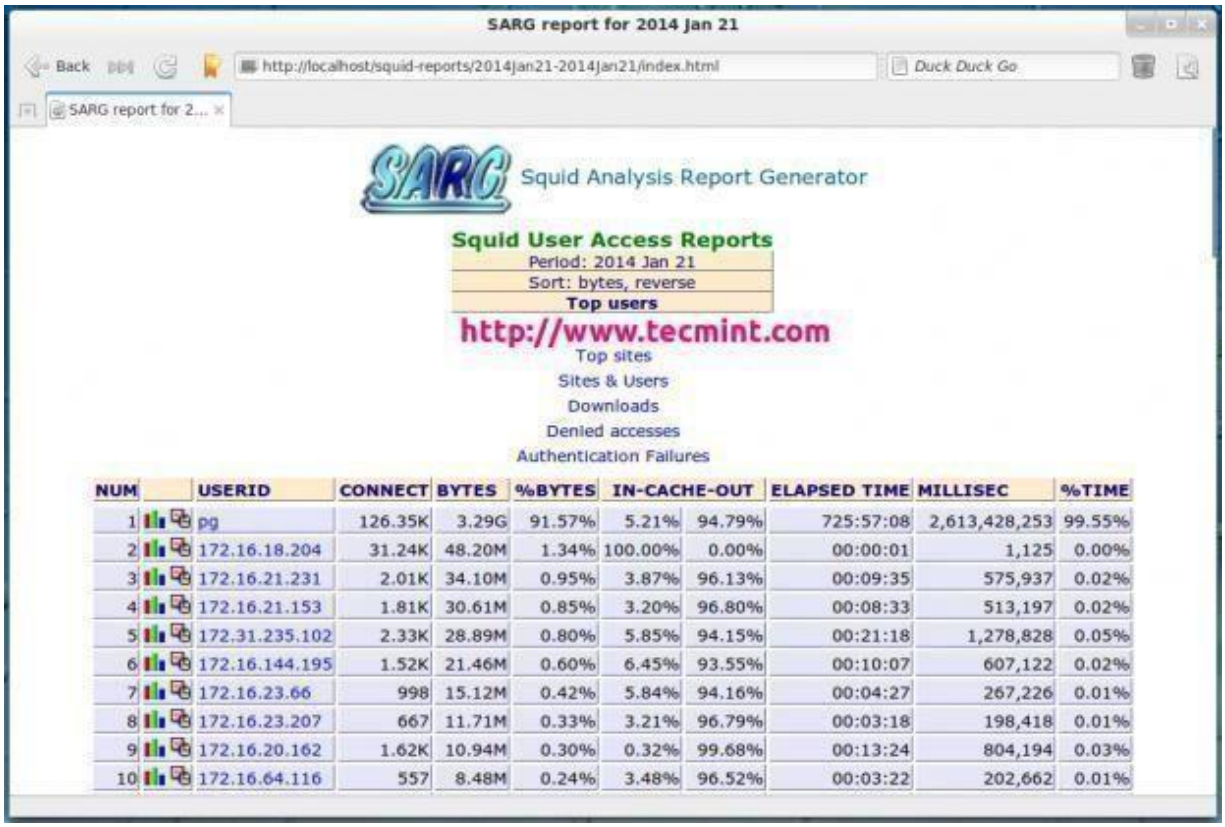
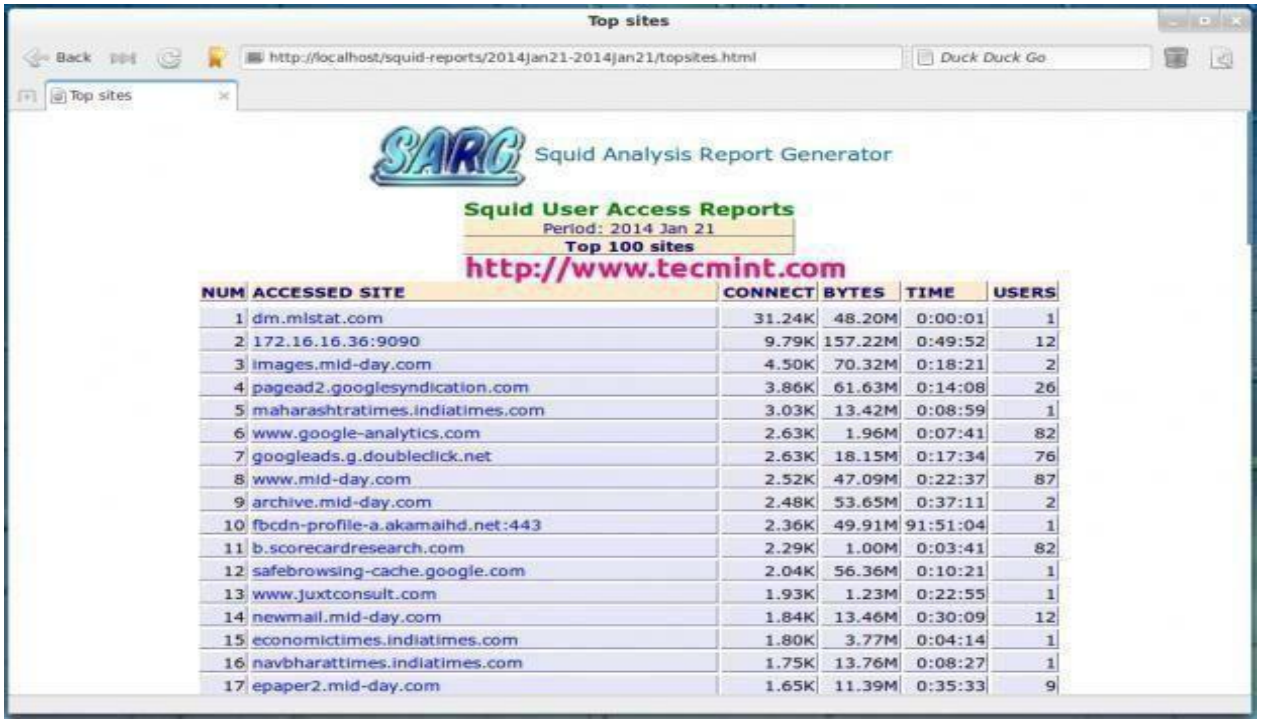


Fig2.User Report



NUM		USERID	CONNECT	BYTES	%BYTES	IN-CACHE-OUT		ELAPSED TIME	MILLISEC	%TIME
1		pg	126.35K	3.29G	91.57%	5.21%	94.79%	725:57:08	2,613,428,253	99.55%
2		172.16.18.204	31.24K	48.20M	1.34%	100.00%	0.00%	00:00:01	1,125	0.00%
3		172.16.21.231	2.01K	34.10M	0.95%	3.87%	96.13%	00:09:35	575,937	0.02%
4		172.16.21.153	1.81K	30.61M	0.85%	3.20%	96.80%	00:08:33	513,197	0.02%
5		172.31.235.102	2.33K	28.89M	0.80%	5.85%	94.15%	00:21:18	1,278,828	0.05%
6		172.16.144.195	1.52K	21.46M	0.60%	6.45%	93.55%	00:10:07	607,122	0.02%
7		172.16.23.66	998	15.12M	0.42%	5.84%	94.16%	00:04:27	267,226	0.01%
8		172.16.23.207	667	11.71M	0.33%	3.21%	96.79%	00:03:18	198,418	0.01%
9		172.16.20.162	1.62K	10.94M	0.30%	0.32%	99.68%	00:13:24	804,194	0.03%
10		172.16.64.116	557	8.48M	0.24%	3.48%	96.52%	00:03:22	202,662	0.01%

Fig 3. Specific Date



NUM	ACCESSED SITE	CONNECT	BYTES	TIME	USERS
1	dm.mistat.com	31.24K	48.20M	0:00:01	1
2	172.16.16.36:9090	9.79K	157.22M	0:49:52	12
3	images.mid-day.com	4.50K	70.32M	0:18:21	2
4	pagead2.googlesyndication.com	3.86K	61.63M	0:14:08	26
5	maharashtratimes.indiatimes.com	3.03K	13.42M	0:08:59	1
6	www.google-analytics.com	2.63K	1.96M	0:07:41	82
7	googleads.g.doubleclick.net	2.63K	18.15M	0:17:34	76
8	www.mid-day.com	2.52K	47.09M	0:22:37	87
9	archive.mid-day.com	2.48K	53.65M	0:37:11	2
10	fbcdn-profile-a.akamaihd.net:443	2.36K	49.91M	91:51:04	1
11	b.scorecardresearch.com	2.29K	1.00M	0:03:41	82
12	safebrowsing-cache.google.com	2.04K	56.36M	0:10:21	1
13	www.juxtconsult.com	1.93K	1.23M	0:22:55	1
14	newmail.mid-day.com	1.84K	13.46M	0:30:09	12
15	economictimes.indiatimes.com	1.80K	3.77M	0:04:14	1
16	navbharattimes.indiatimes.com	1.75K	13.76M	0:08:27	1
17	epaper2.mid-day.com	1.65K	11.39M	0:35:33	9

Fig 4. Top Accessed Sites

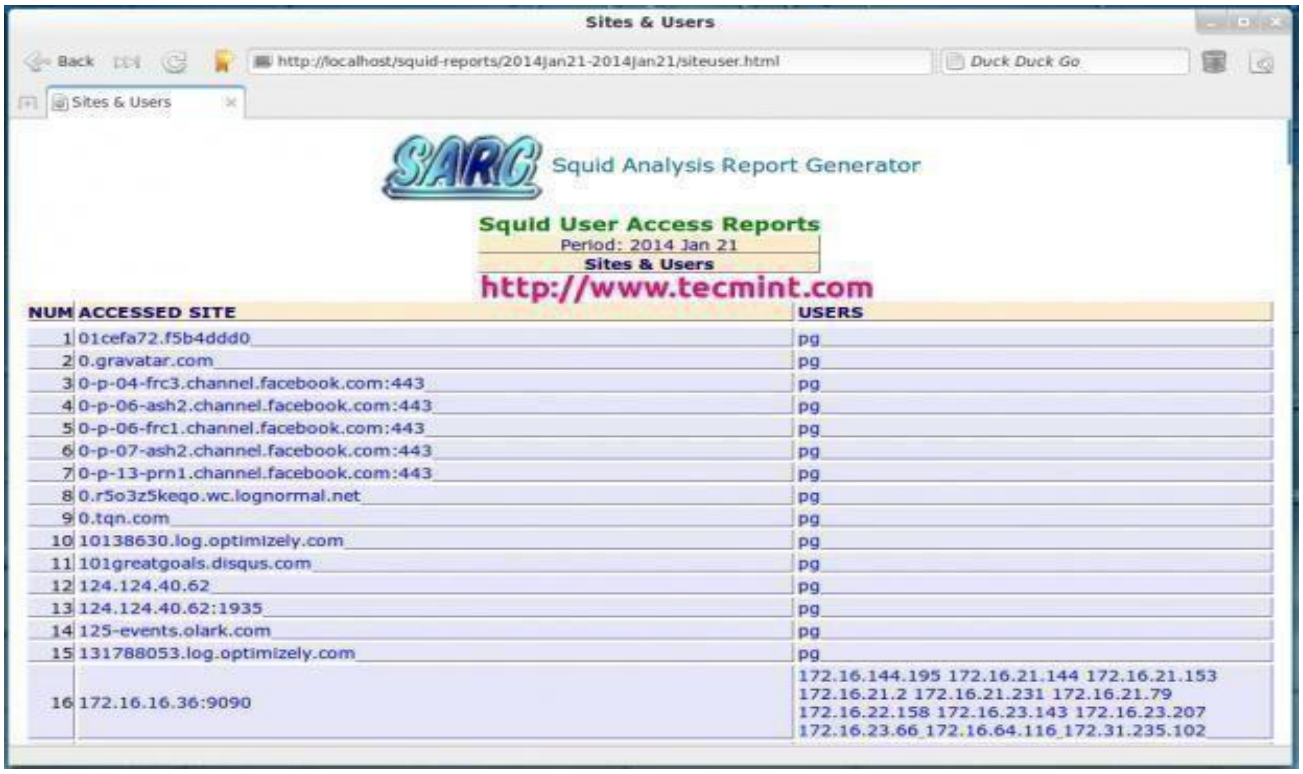


Fig 5. Top Sites and Users

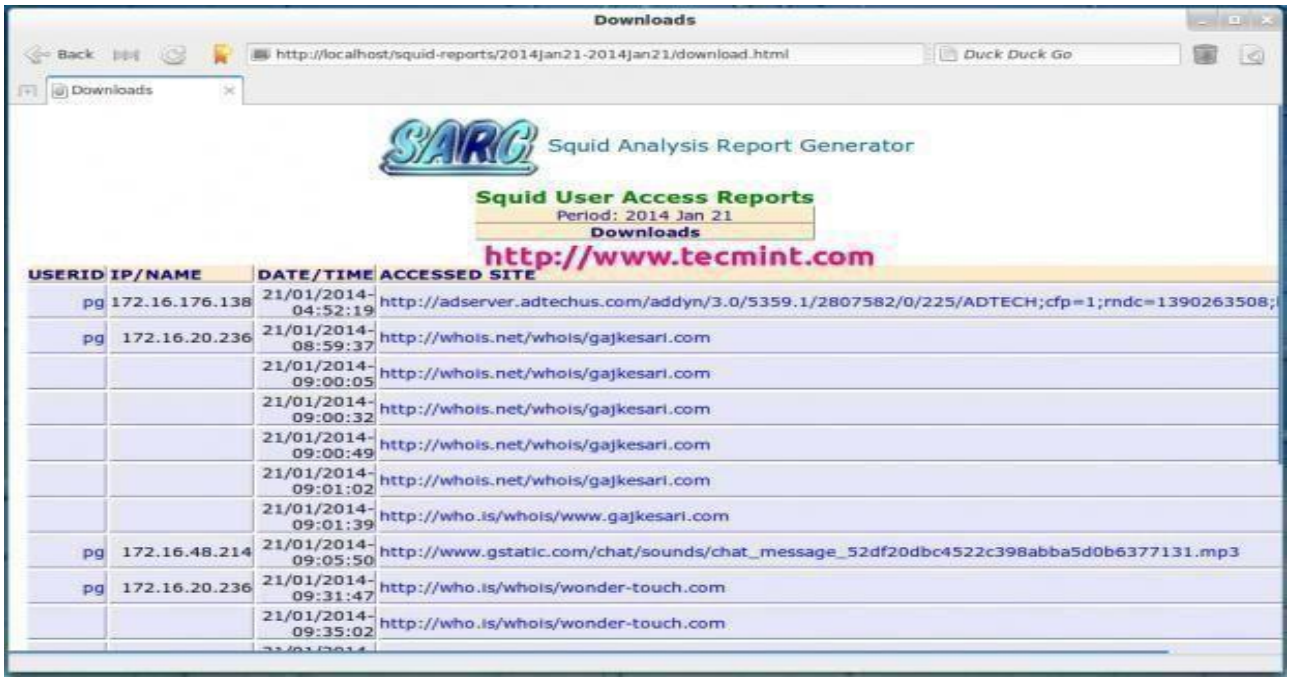


Fig 6. Top Downloads

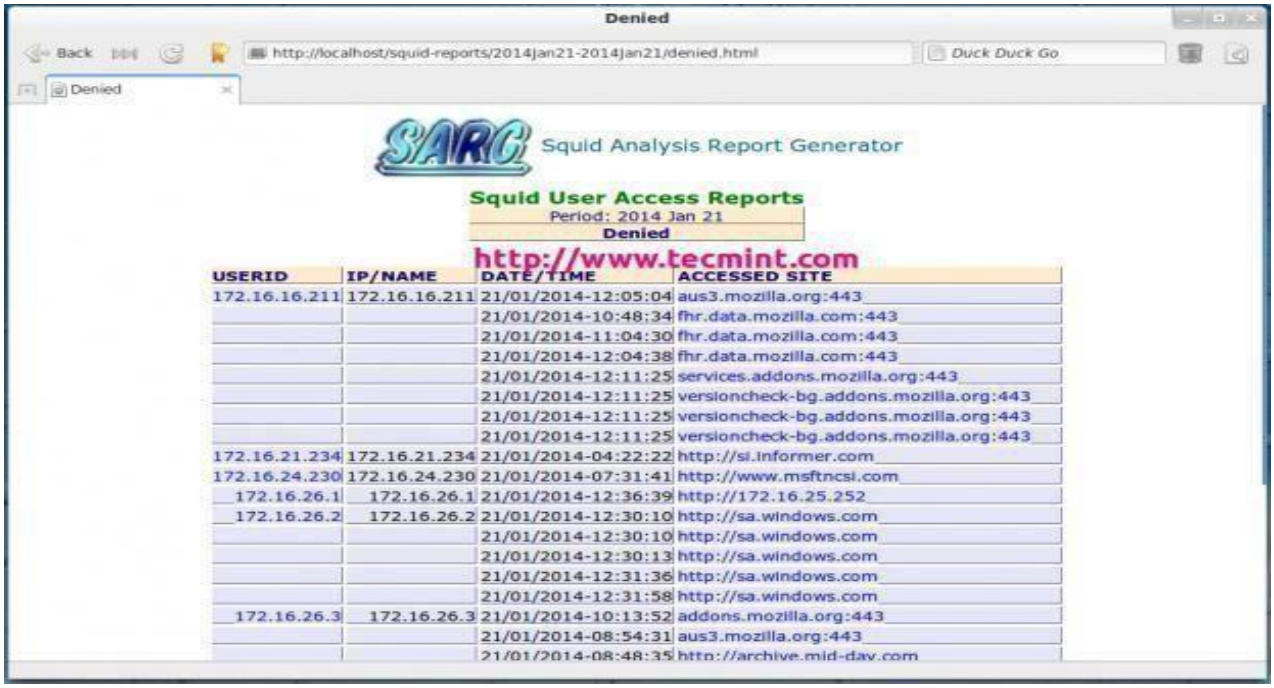


Fig 7. Denied Access

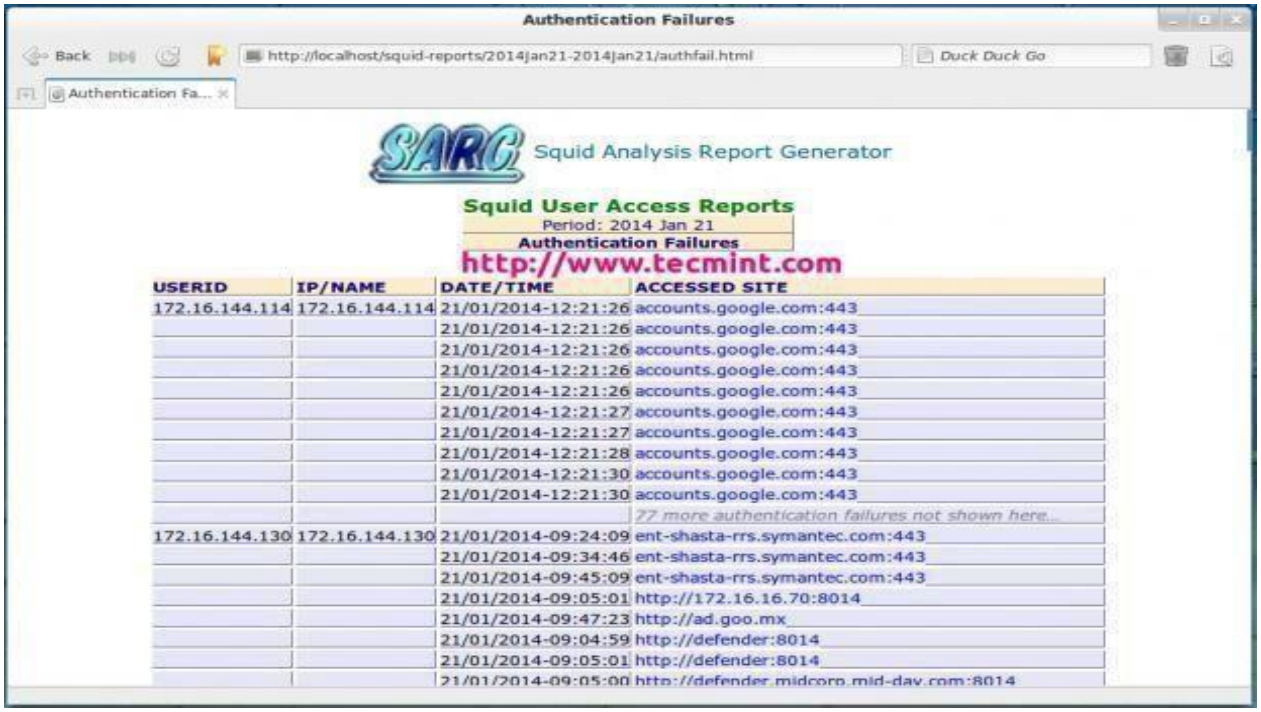


Fig 8. Authentication Failures